

Finite-Cell Gate Derivations and Remaining Operator Targets: Surface Spectrum, Laplace Matching, GP/NLS Shell Response, and Hodge Capacity

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Abstract

The companion papers isolate a dimensionless fine-structure-scale closure based on the ideal-trefoil ropelength, a spherical pressure cell, and a finite-cell BEM/NLS stationary point. This note records the coefficient-gate status after the post-review calculations. The pressure-sector count $N_p = 4$ is derived from the fixed-volume surface spectrum and is stable under perturbative, nonlinear star-shaped, and finite-perimeter global-topology checks. The Laplace-matched reciprocal pressure model gives $\chi_R = 2$ within the minimal reciprocal action, provided same-order nonreciprocal terms are excluded. The GP/NLS shell audit derives

$$\sigma = 3 + \frac{2}{3}w_\perp,$$

but the precision value $11/48$ follows only after imposing $w_\perp = 1$; the current relaxed-core proxy returns $w_\perp \simeq 2.076$, so this remains an open primitive gate. The exterior Hodge calculation gives $q_\phi = 1$ and $\Lambda_\phi = 4\pi R\mathcal{K}_{\text{cell}}$, while the canonical normal-mode reduction gives $\Lambda_\phi = E_{\text{eff}}/2$ after the stated E_{eff} -bridge. The package is therefore a conditional finite-cell derivation/certificate, not a complete first-principles derivation of the QED fine-structure constant.

1 Purpose and status

The preceding companion papers separate three levels of claim. The first is an algebraic closure audit. The second is a spherical/NLS pressure-cell construction. The third is a finite-cell BEM/NLS stationary-point calculation and conditional far-field consistency test. The present note supplies the missing logical bridge between a free closure model and a derived effective model: it proves uniqueness of the minimal finite-cell variational reduction under explicitly stated axioms.

The status of the result is therefore

derived within the unique minimal finite-cell variational reduction.

It is not, by itself,

derived from the full microscopic dynamics without reduction assumptions.

That stronger statement would require deriving the axioms below directly from the underlying field theory.

2 Post-review gate status

The achieved results are:

$N_p = 4$ [derived in spherical surface spectrum; stable under perturbative, nonlinear star-shaped, and finite-p

$$A_\chi = \chi_R + \frac{N_p}{\chi_R}, \quad \chi_R = 2 \quad [\text{derived in Laplace-matched reciprocal pressure model}],$$

$$\frac{11}{48} \quad [\text{derived within unweighted isotropic GP/NLS shell reduction}],$$

and

$$q_\phi = 1, \quad \Lambda_\phi = 4\pi R K_{\text{cell}} \quad [\text{derived by exterior Hodge capacity}].$$

The remaining open targets are:

$$w_\perp = 1 \quad [\text{full microscopic GP/SST core reduction}],$$

and

$$\Lambda_\phi = \frac{E_{\text{eff}}}{2} \quad [\text{interior one-cell phase Hessian}].$$

The earlier delay-stability route is no longer used as the primary justification for $N_p = 4$, because its default parameters did not cleanly retain the $\ell = 1$ sector. The surface-spectrum route is parameter-free at quadratic order and is therefore the preferred derivation.

3 Axioms of the minimal finite-cell reduction

Let the finite cell be represented by a spherical boundary S^2 , with outward unit normal $\mathbf{n}$, pressure field p , dimensionless shell thickness $\eta = a/R_{\text{cell}}$, cell-radius ratio $\chi_R = R_{\text{cell}}/L_K$, and exterior phase field u . The minimal quadratic reduction is assumed to satisfy:

- A1. $SO(3)$ covariance on the spherical boundary.
- A2. Locality at quadratic order on S^2 .
- A3. Positive quadratic energy for stable boundary perturbations.
- A4. Pressure constraints only for total compression and centre displacement:

$$\int_{S^2} p \, d\Omega = P_0, \quad \int_{S^2} p \mathbf{n} \, d\Omega = \mathbf{P}_1.$$

- A5. Isotropic NLS/GP shell-gradient response at second order.
- A6. Reciprocal inner/outer pressure balance at leading scale order.
- A7. Leading far-field coupling carried only by the exterior $1/r$ harmonic phase mode.

These assumptions define the class in which the uniqueness statement is made.

4 The finite-cell effective action

At quadratic order the effective action separates into

$$\mathcal{A}_{\text{eff}} = \mathcal{A}_p[p] + \mathcal{A}_{\text{shell}}[\eta] + A_\chi(\chi_R) + \mathcal{A}_\phi[u].$$

The four pieces are fixed below by the axioms, up to an overall normalization that cancels in the dimensionless coefficient gates.

5 Theorem 1: constrained pressure minimization gives $N_p = 4$

Consider

$$\mathcal{A}_p[p] = \frac{1}{2} \int_{S^2} |\nabla_{S^2} p|^2 d\Omega$$

subject to

$$\int_{S^2} p d\Omega = P_0, \quad \int_{S^2} p \mathbf{n} d\Omega = \mathbf{P}_1.$$

Introducing Lagrange multipliers λ_0 and $\boldsymbol{\lambda}_1$, the Euler–Lagrange equation is

$$-\Delta_{S^2} p = \lambda_0 + \boldsymbol{\lambda}_1 \cdot \mathbf{n}.$$

The constant function belongs to $\mathcal{H}_0$, and the coordinate functions n_x, n_y, n_z span $\mathcal{H}_1$. Hence the minimal pressure subspace is

$$\mathcal{P}_{\min} = \mathcal{H}_0 \oplus \mathcal{H}_1.$$

Since

$$\dim \mathcal{H}_\ell = 2\ell + 1,$$

one obtains

$$N_p = \dim \mathcal{H}_0 + \dim \mathcal{H}_1 = 1 + 3 = 4.$$

This proves uniqueness of $N_p = 4$ inside the lowest $SO(3)$ -closed pressure subspace satisfying the volume and centre-displacement constraints. Modes $\ell \geq 2$ represent quadrupolar and higher shape deformations and are not required by the constraints.

6 Theorem 1b: fixed-volume surface spectrum gives the same $N_p = 4$

The constrained pressure argument above is strengthened by the fixed-volume surface spectrum. Let the spherical boundary be

$$r(\Omega) = R[1 + \epsilon f(\Omega)].$$

At fixed volume,

$$\delta^2 A = \frac{R^2}{2} \int_{S^2} (|\nabla_{S^2} f|^2 - 2f^2) d\Omega.$$

For $f = Y_{\ell m}$,

$$k_\ell = \ell(\ell + 1) - 2 = (\ell - 1)(\ell + 2).$$

Thus $k_1 = 0$, identifying the three translation/centre-displacement modes. For $\ell \geq 2$, $k_\ell > 0$, so these are positive-stiffness shape modes integrated out at zeroth order. The $\ell = 0$ mode is retained separately as the scalar compression degree of freedom. Hence

$$\mathcal{P}_{\min} = \mathcal{H}_0 \oplus \mathcal{H}_1, \quad N_p = 1 + 3 = 4.$$

This is the preferred derivation of $N_p = 4$ in the present package.

7 Theorem 1c: finite-core and nonlinear $\ell \geq 2$ shape stability

The first positive fixed-volume shape-mode gap is

$$k_2 = 4.$$

Let δH be the finite-core or weak nonspherical perturbation of the $\ell \geq 2$ shape Hessian. If

$$\|\delta H\|_{\text{op}} < \frac{k_2}{2} = 2,$$

then the $\ell \geq 2$ block remains separated from the retained $\mathcal{H}_0 \oplus \mathcal{H}_1$ pressure manifold. The finite-core scale in the trefoil cell is

$$\eta_K = \frac{1}{4\mathcal{L}_K} = 1.52703116982 \times 10^{-2},$$

far below this half-gap bound.

A nonlinear star-shaped audit tests finite-amplitude deformations

$$r(\Omega) = \exp\left(\sum_{\ell=2}^{\ell_{\max}} \sum_m a_{\ell m} Y_{\ell m}(\Omega)\right)$$

with exact volume rescaling to $V = 4\pi/3$. For $\ell_{\max} = 6$, the $\ell \geq 2$ coefficient space has 45 real modes. Across 300 random samples per amplitude at $\eta_K, 0.05, 0.10, 0.25, 0.50, 1.00$, no negative fixed-volume area excess was found. At the physical scale,

$$\min \Delta A_{\text{random}} = 1.71825494947 \times 10^{-3}.$$

Moreover, 20 L-BFGS-B local minimizations from amplitude-one random starts returned to the spherical solution, with

$$\Delta A_{\text{final}} \lesssim 1.2 \times 10^{-12}, \quad \max \|a_{\ell m}^{\text{final}}\| = 5.830157 \times 10^{-7}.$$

Thus nonlinear $\ell \geq 2$ star-shaped deformations do not enter the leading pressure manifold in the tested class.

8 Theorem 1d: global topology admissibility

The nonlinear audit covers star-shaped surfaces. The global loophole is closed for the larger admissible class of bounded finite-perimeter pressure cells by the Euclidean isoperimetric inequality:

$$P(\Omega)^3 \geq 36\pi V(\Omega)^2,$$

where $P(\Omega)$ is the perimeter and $V(\Omega)$ the enclosed volume. Equality holds only for a ball, up to null sets. Hence, at fixed $V(\Omega) = 4\pi R^3/3$,

$$P(\Omega) \geq 4\pi R^2, \quad \Delta A = P(\Omega) - 4\pi R^2 \geq 0.$$

Therefore no admissible non-star-shaped, disconnected, or higher-genus finite-perimeter pressure cell can lower the fixed-volume area below the sphere. Self-intersecting immersed surfaces are not admissible reduced boundaries of a physical pressure cell; if counted with multiplicity, their area cannot improve the isoperimetric bound. Thus global topology does not promote $\ell \geq 2$ modes into the leading pressure manifold.

9 Theorem 2: isotropic shell normalization gives $\sigma = 11/3$ only if $w_{\perp} = 1$

Let

$$\eta = \frac{a}{R_{\text{cell}}}.$$

The volume Jacobian for a relative inward shell displacement is

$$\frac{V(R-a)}{V(R)} = (1-\eta)^3 = 1 - 3\eta + 3\eta^2 + O(\eta^3).$$

After the first-order displacement is absorbed into the equilibrium radius, the second-order volume contribution is

$$\sigma_{\text{vol}} = 3.$$

The isotropic NLS/GP gradient term contributes through the transverse projector

$$P_{\perp} = I - \hat{\mathbf{r}}\hat{\mathbf{r}}.$$

Using

$$\langle \hat{r}_i \hat{r}_j \rangle_{S^2} = \frac{1}{3} \delta_{ij},$$

one gets

$$\langle P_{\perp} \rangle_{S^2} = \frac{2}{3} I.$$

Equivalently,

$$\langle \sin^2 \theta \rangle_{S^2} = \frac{\int_0^{\pi} \sin^2 \theta \sin \theta d\theta}{\int_0^{\pi} \sin \theta d\theta} = \frac{2}{3}.$$

Thus the second-order shell coefficient has the form

$$\sigma = \sigma_{\text{vol}} + \sigma_{\perp} = 3 + \frac{2}{3} w_{\perp}.$$

Only for the unweighted isotropic shell normalization $w_{\perp} = 1$ does this reduce to

$$\sigma = \frac{11}{3}.$$

The GP core-profile audit confirms the angular result and computes the radial profile. It shows that the normalized angular factor $2/3$ is profile independent for isotropic radial weights. It also shows that alternative profile-weight proxies do not generically reproduce $11/48$. More sharply, the relaxed GP-core Hessian proxy gives $w_{\perp} \simeq 2.076$, not 1. Therefore the equality $w_{\perp} = 1$ is not derived by the current primitive GP/NLS tests; it is the remaining microscopic normalization gate.

10 Theorem 3: reciprocal pressure stationarity gives $\chi_R = 2$

The equal inner/outer normalization is now supported by the SST pressure identity. With the canonical constants,

$$P_{\text{kin}} = \frac{1}{2} \rho_{\text{core}} \|\mathbf{v}_{\odot}\|^2 = 2.329244600448 \times 10^{30} \text{ Pa},$$

and

$$P_{\text{Laplace}} = \frac{F_{\text{swirl}}^{\text{max}}}{2\pi r_c^2} = 2.329244600448 \times 10^{30} \text{ Pa}.$$

Therefore the leading pressure ratio is

$$\mu = \frac{P_{\text{in}}}{P_{\text{out}}} = 1.$$

The general reciprocal action

$$A_{\chi} = \chi_R + \mu \frac{N_p}{\chi_R}$$

then reduces to the self-dual form used below.

Let

$$\chi_R = \frac{R_{\text{cell}}}{L_K}.$$

The leading outer pressure-extension cost is proportional to χ_R , while the inner crowding/compliance cost is reciprocal and proportional to the number of pressure sectors. Inner/outer reciprocity therefore fixes the minimal scale action to

$$A_\chi(\chi_R) = \chi_R + \frac{N_p}{\chi_R}.$$

Stationarity gives

$$\frac{dA_\chi}{d\chi_R} = 1 - \frac{N_p}{\chi_R^2} = 0,$$

so

$$\chi_R = \sqrt{N_p} = 2.$$

Equivalently, if

$$A_\chi = \chi_R + \frac{\lambda_\chi}{\chi_R},$$

then

$$\lambda_\chi = N_p = 4.$$

This is unique in the leading-order class of reciprocal, scale-homogeneous inner/outer pressure actions. If additional terms such as χ_R^2 , χ_R^{-2} , or $\log \chi_R$ are admitted at the same order, uniqueness is lost; such terms are excluded by the minimal scale-reciprocity axiom.

11 Theorem 4: the exterior phase Hessian fixes $q_\phi = 1$

An exterior harmonic field has the multipole expansion

$$u(r, \theta, \phi) = \sum_{\ell, m} a_{\ell m} \frac{Y_{\ell m}(\theta, \phi)}{r^{\ell+1}}.$$

Only $\ell = 0$ gives a $1/r$ field. The $\ell = 1$ sector begins at r^{-2} , and all $\ell \geq 2$ modes are shorter-ranged. Therefore the leading far-field charge is the constant boundary phase, i.e. the unique $H^0(S^2)$ mode. Since S^2 is connected,

$$q_\phi = \dim H^0(S^2) = 1.$$

For the exterior monopole

$$u(r) = \frac{\phi}{r},$$

one has

$$\int_{r \geq 1} |\nabla u|^2 d^3x = 4\pi\phi^2.$$

Thus

$$\Lambda_\phi = 4\pi\mathcal{K}_{\text{cell}}.$$

12 Theorem 4b: canonical phase–pressure normal modes give $\Lambda_\phi = E_{\text{eff}}/2$

The remaining one-cell Hessian is supplied by the canonical linearized Madelung/GP normal-mode reduction. The density perturbation and phase are a canonical pair; after diagonalization each interior mode has Hamiltonian

$$H_k = \frac{1}{2}p_k^2 + \frac{1}{2}\omega_k^2 q_k^2.$$

For a periodic normal mode,

$$\left\langle \frac{1}{2}p_k^2 \right\rangle_T = \left\langle \frac{1}{2}\omega_k^2 q_k^2 \right\rangle_T.$$

The phase/flow and pressure/compression budgets therefore split equally:

$$M_\phi = M_p = \frac{E_{\text{eff}}}{2}.$$

Combining this half-budget identity with the accessible-area projection and the radial optimum gives

$$\Lambda_\phi = \frac{E_{\text{eff}}}{2R^2}.$$

For the normalized finite cell $R = 1$,

$$\Lambda_\phi = \frac{E_{\text{eff}}}{2},$$

and therefore, from Theorem 4,

$$\mathcal{K}_{\text{cell}} = \frac{E_{\text{eff}}}{8\pi}.$$

The remaining physical identification is that the E_{eff} used in the pressure-cell chain is the cycle-averaged quadratic energy of this same canonical interior mode family.

13 Resulting pressure scale

Combining Theorems 1–3 gives a conditional pressure scale. The derived sector count $N_p = 4$ must be supplemented by the sector-volume normalization $4\pi/3$ per pressure sector per unit ropelength:

$$E_p^{(0)} = N_p \frac{4\pi}{3} \mathcal{L}_K = \frac{16\pi}{3} \mathcal{L}_K.$$

This $4\pi/3$ -per-sector normalization is a load-bearing closure coefficient, not yet a variational theorem. Moreover,

$$\eta_K = \frac{1}{2\chi_R \mathcal{L}_K} = \frac{1}{4\mathcal{L}_K}.$$

With the additional normalization $w_\perp = 1$, one has $\sigma = 11/3$, and hence

$$\sigma \eta_K^2 = \frac{11}{3} \frac{1}{16\mathcal{L}_K^2} = \frac{11}{48\mathcal{L}_K^2}.$$

If $w_\perp \neq 1$, this precision correction changes. Therefore

$$E_p^{\text{NLS}} = \frac{16\pi}{3} \mathcal{L}_K \left(1 - \frac{11}{48\mathcal{L}_K^2} \right).$$

Theorem 4 then gives the far-field stiffness relation inside the canonical normal-mode reduction,

$$\mathcal{K}_{\text{cell}} = \frac{E_{\text{eff}}}{8\pi},$$

subject to the bridge identifying E_{eff} with the cycle-averaged quadratic energy of the same interior mode family. Only after the physical far-field law is established does the dimensionless cell coefficient

$$\alpha_{\text{cell}} = \frac{1}{4\pi\mathcal{K}_{\text{cell}}} = \frac{2}{E_{\text{eff}}}$$

become an electromagnetic coupling rather than a fine-structure-scale cell number.

14 Reproducibility map

The supporting scripts and output files are stored in `../code/` relative to the `Manuscripts` directory:

- `derive_pressure_mode_cutoff_surface_spectrum.py`: linear surface-spectrum derivation of $N_p = 4$;
- `test_finite_core_nonspherical_shape_corrections.py`: perturbative finite-core/nonspherical half-gap audit;
- `solve_nonlinear_shape_stability.py`: nonlinear star-shaped $\ell \geq 2$ shape-stability solve;
- `derive_gp_core_profile_second_variation.py`: GP/NLS shell coefficient and $w_{\perp}$ sensitivity audit;
- `derive_pressure_self_duality_from_laplace_matching.py`: Laplace-matched reciprocal radius action;
- `solve_one_cell_hodge_phase_hessian.py`: exterior Hodge capacity and $q_{\phi} = 1$;
- `derive_phase_pressure_exchange_self_duality.py`: canonical normal-mode derivation of $M_{\phi} = M_p = E_{\text{eff}}/2$ and $\Lambda_{\phi} = E_{\text{eff}}/2$.

The numerical CSV and Markdown outputs are stored in the corresponding `../code/outputs_*` directories and summarized in `../code/DERIVED_GATE_EVIDENCE_MANIFEST.json`.

15 Falsification and robustness checks

The uniqueness theorem depends on the minimal axioms. The following checks test those axioms rather than the algebra:

1. include the $\ell = 2$ pressure sector and verify that it enters only as a higher-order shape correction;
2. replace $P_{\perp}$ by a weighted transverse projector and verify that isotropic NLS/GP reduction fixes the weight to unity;
3. replace $A_{\chi} = \chi_R + N_p/\chi_R$ by $a\chi_R + bN_p/\chi_R$ and verify that inner/outer reciprocity enforces $a = b$;
4. compute the one-cell phase Hessian directly from the full finite-cell operator and verify $\Lambda_{\phi} = E_{\text{eff}}/2$.

16 Conclusion

The four gate calculations now have differentiated status. The pressure-sector count is derived by the spherical fixed-volume surface spectrum and is stable under perturbative finite-core, nonlinear star-shaped, and global finite-perimeter topology checks:

$$k_\ell = (\ell - 1)(\ell + 2), \quad \mathcal{P}_{\min} = \mathcal{H}_0 \oplus \mathcal{H}_1, \quad N_p = 4.$$

The radius action is derived within the Laplace-matched reciprocal pressure model because the SST kinetic and circular Laplace pressures agree, giving $\mu = 1$, $A_\chi = \chi_R + N_p/\chi_R$, and $\chi_R = 2$. The finite-shell coefficient $11/48$ is conditional on the unweighted isotropic GP/NLS shell normalization $w_\perp = 1$; the primitive-equation derivation remains open and the current relaxed-core proxy gives $w_\perp \simeq 2.076$. The exterior phase-Hodge calculation derives $q_\phi = 1$ and $\Lambda_\phi = 4\pi R\mathcal{K}_{\text{cell}}$. The interior Hessian identity $\Lambda_\phi = E_{\text{eff}}/2$ is derived within the canonical linearized phase-pressure normal-mode reduction, provided E_{eff} is identified with the cycle-averaged quadratic energy of that same interior mode family.

Thus the package has moved beyond a free closure model for N_p , χ_R , and the far-field phase stiffness. It has isolated $11/48$ to a single microscopic normalization gate and has isolated the far-field bridge to one physical identification of E_{eff} . A complete first-principles derivation of the electromagnetic fine-structure constant still requires

$$w_\perp = 1$$

from the full GP/SST core-profile reduction and the identification of E_{eff} with the cycle-averaged quadratic energy of the canonical interior mode family.

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